Visual System one of the most important one?

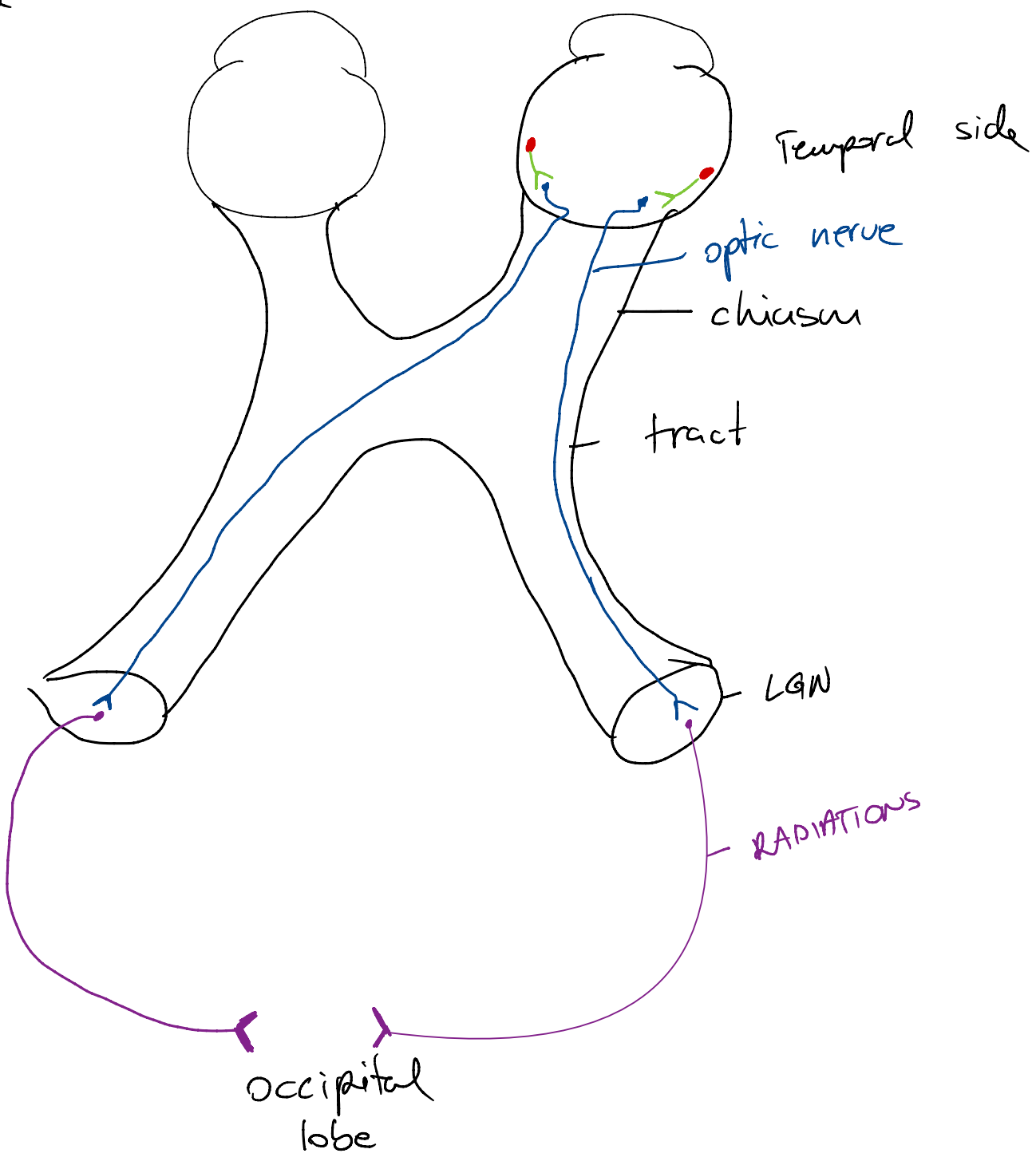
Images used: 15.8, 15.12, 15.13, 15.15, 15.14, 15.6, 15.9

I. Overview of retinogeniculocortical pathway

retina — lateral geniculate nucleus — brain

ganglion cells are very long

• photoreceptor to bipolar cells to



A. Optical system

B. Retina

C. Optic nerve

D. Chiasm

E. Optic tract

F. Lateral geniculate nucleus

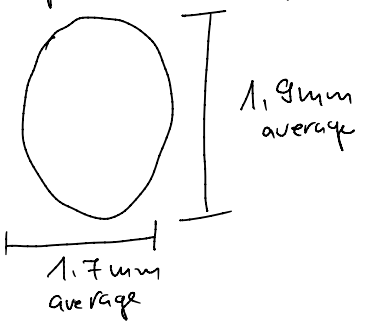
G. Optic radiations

H. Visual cortex

II. Optic nerve (cranial nerve II) ca. 5cm long and have 4 sections.

A. Four sections

1. Intraocular - optic disc / optic nerve head



a. Ganglion cells travel from all over the retina to converge on the optic disc.

b. Physiologic cup area where we do not have ganglion cells

- depression in center
- whiter
- in glaucoma cup gets bigger
- Bipolar cells and no photoreceptors

c. Blind spot

No photoreceptors that is the cause for blind spot

- 15 degrees temporal
- slightly bellow fixation

2. Intraorbital portion that is in the orbit behind the eyeball. Posses rectus muscles. Fibers are very organized

a. Once the optic nerve leaves the globe, it is surrounded by pia, arachnoid, and dura around the brain

b. Clinical correlate: Papilledema

are the margins: pink + distict can I make edges more red - indicate swealing

c. Posterior to the globe, the fibers become myelinated by oligodendrocytes. central nervous system

MS attackt oligodendrocytes

Patients will get optic neuritis

3. Intracanalicular part that is in the optic canal
   sphenoid bone - body + LW (AI check)

4. Intracranial
   bones for our head?

III. Optic chiasm important to know what is around that

A. Neighboring structures

circule of Willis

ICA (?? artery)

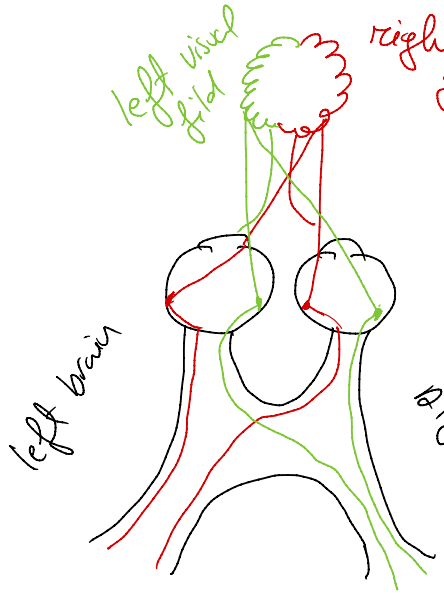
1cm bellow chicom we will find pituitary gland

B. Arrangement of fibers

temporal fibers → ipsilateral

nasal fibers → contralat

everything on the left visual field ended up on the RIGHT side of the brain - Because of CHIAZM ?



everything on the right visual field ended up on the left side of the brain.

IV. Optic tract

A. Most ganglion cell fibers travel through the optic tract to the LGN

B. Some ganglion cell fibers leave the optic tract prior to reaching the LGN

1. Superior colliculi

reflex head + saccadic mvt.

2. Suprachiasmatic nuclei

circadian rhythm

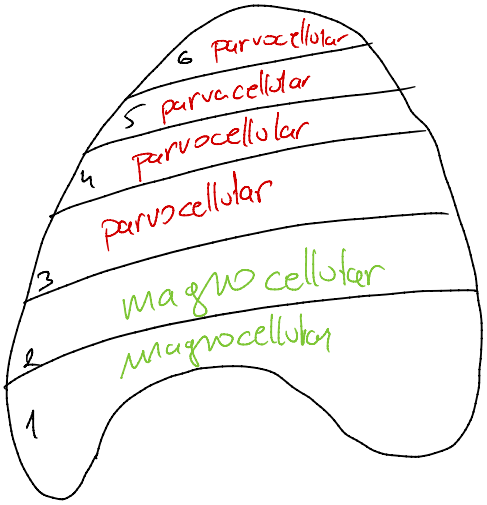
3. Pretectum

pupils

V. Lateral geniculate nuclei that will allow us to see

A. Two main types of ganglion cells

1. P cells (parvocellular pathway)
   - visual acuity
   - color vision
   - concrat



2. M cells (magnocellular pathway)

- detection of movement
- will let us see the flickering of light

3. Koniocellular cell
   is for blue and yellow color

VI. Optic radiations very spread at through over brain

A. Parietal lobe

fibers from superior retina

B. Temporal lobe

fibers from inferior retina

VII. Visual cortex

A. aka primary visual cortex, V1, or striate cortex

located on midial surface of occ ?? lobe

B. Calcarine fissure

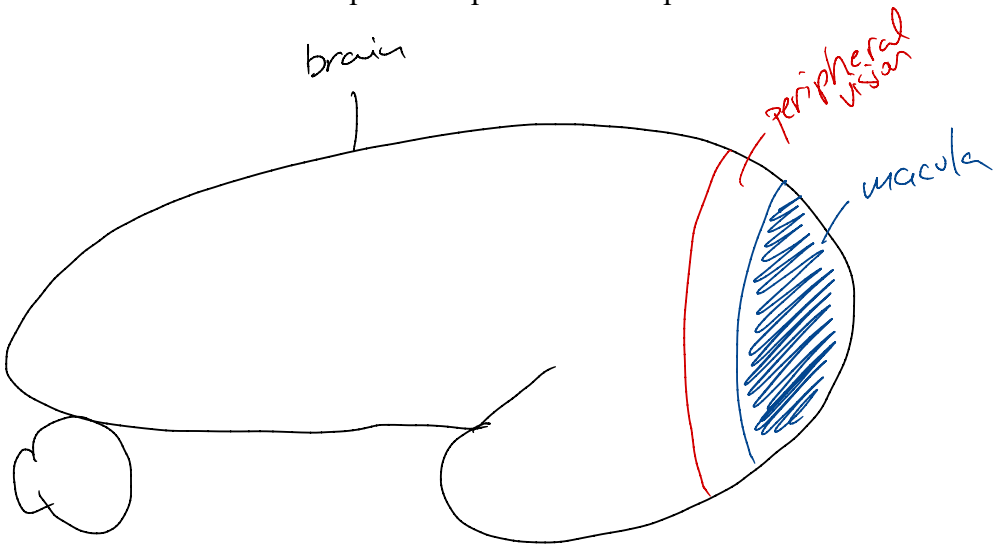
C. Cuneus gyrus

above calcarine fissure
sup. retina

D. Lingual gyrus

below calcarine fissure
inf. retina

E. Macular fibers terminate in the most posterior part of the occipital lobe.



VIII. Clinical implications

A. Information from the right visual field will project to the left visual cortex, and information from the left visual field will project to the right visual cortex.

B. General visual field concepts

1. On the retina, the optic nerve is always more nasal to the macula

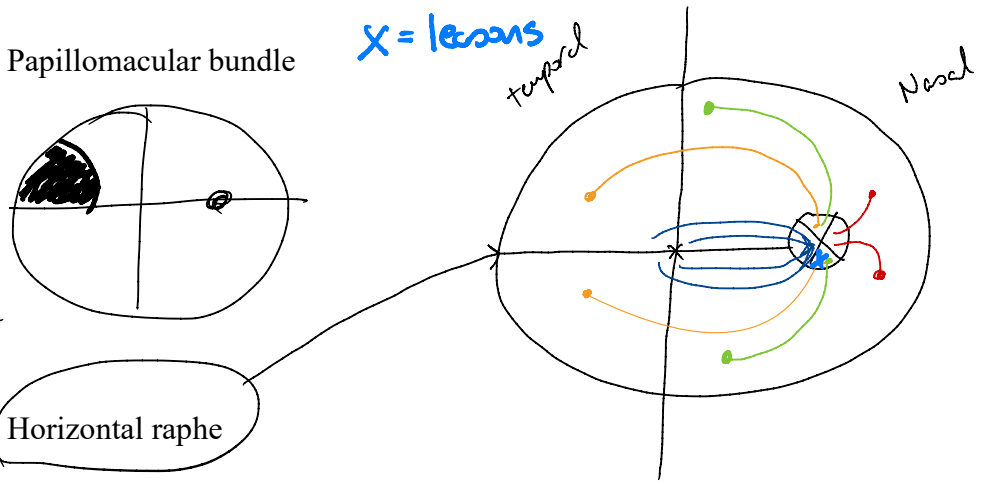
2. Aligning visual fields

3. Position of the blind spot

C. Optic disc

1. Papillomacular bundle

X = lesions



- Temporal retinal fibers
- Macula fibers
- nasal to macula & disc
- nasal to macula temporal to disc

should distinct — Type of visual field: NASAL STEP

X = inferior disc

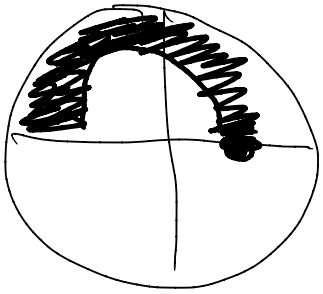
2. Horizontal raphe

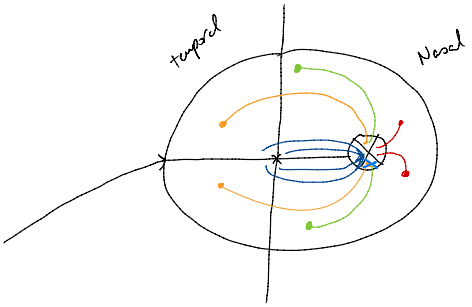
more severe glaucoma

X = temporal fib. + nasal fibers

Type of visual field:

ARCUTE DEFECT (binocular)



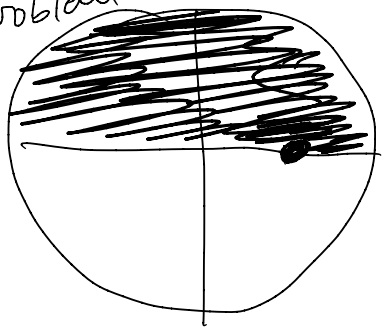


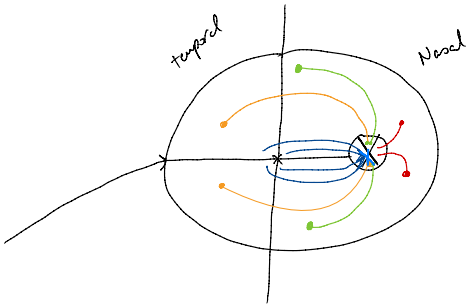
respects horizontal line

Blood vessel problem

X = nasal f. + temp f. + macula f.

Type of VF: ALTITUDINAL DEFECT (monocular)

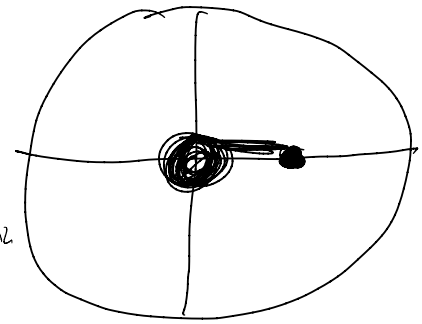


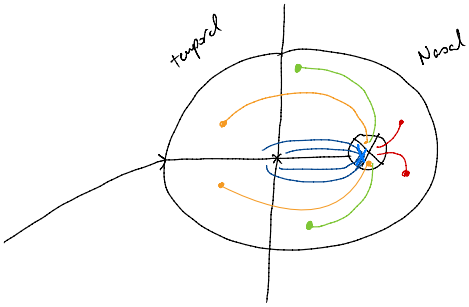


3. Visual field characteristics

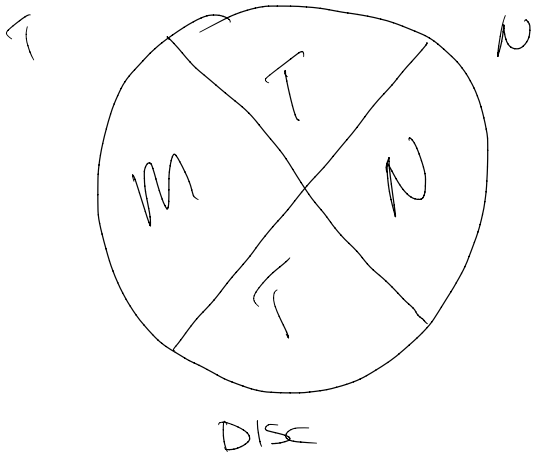
X = only maccula bundl defect

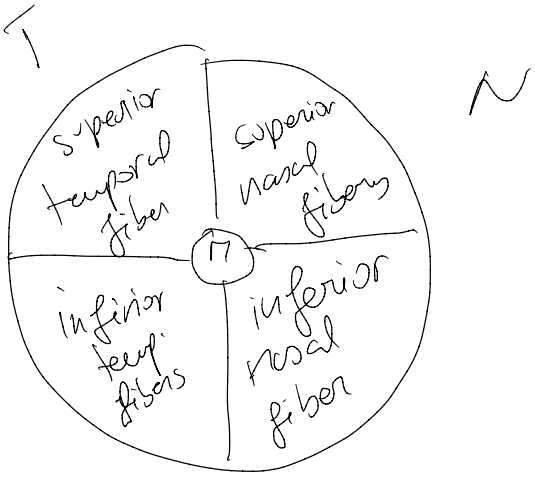
Type of VF: CENTRAL CECOCENTRAL DEFECT





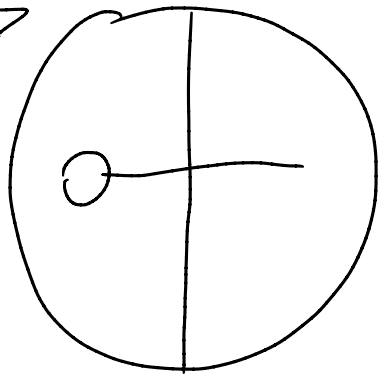
D. Optic nerve (intraorbital, intracanalicular, intracranial)

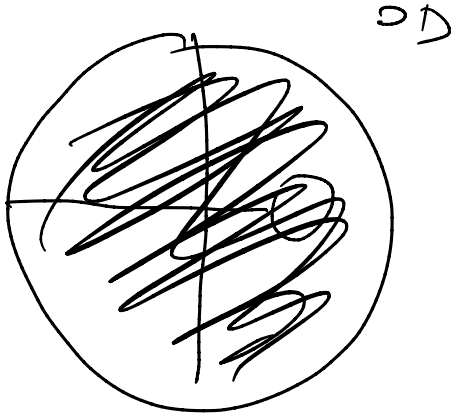




1. Visual field characteristics

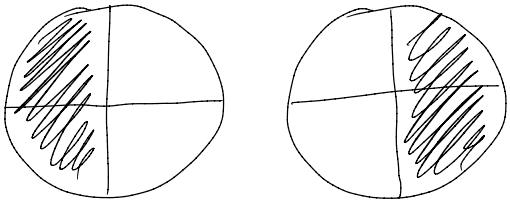
Monocular
Should not respect horizontal line





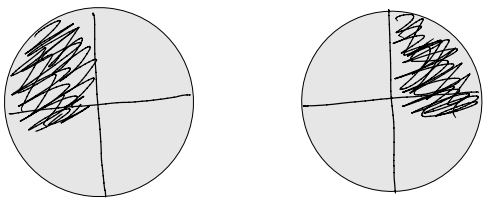
E. Chiasm (perpeti ——) tumor
At check

Temporal defect in both eyes
Nasal retinal fibers
we will have?
coming from both eye

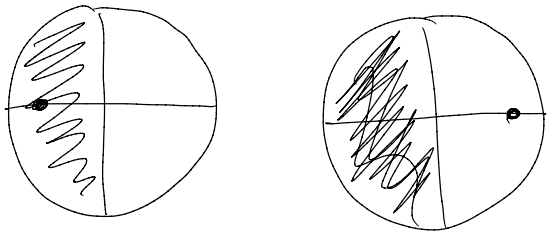


1. Visual field characteristics

opposite site HETRONYMOUS HEMIANOPIA? prevent



F. Optic tract Anything BEHIND CHIAZTA BOTH EYES WILL BE AFFECTED!



1. Visual field characteristics

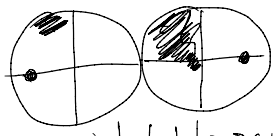
respects vertical line
both eyes
homonymous
↳ same side of the visual field
hemianopia
↳ whole

atrophy of optic nerve → show up like a pallor (whitish/pale)

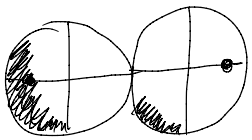
RAPD - if we

G. Radiations spread out fibers

temporal lobe → inf retina → superior VF



parietal lobe → sup retina → inf. VF



1. Visual field characteristics

homonopies hemianopia

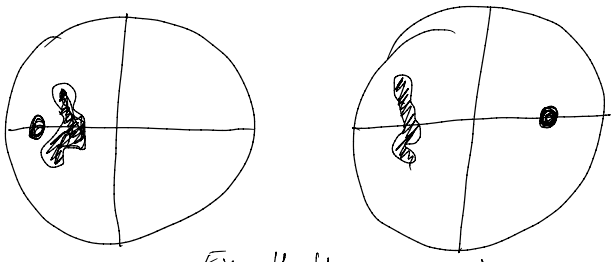
our lesion is gonna be incongruent (diffrent VF deffets in L & R sides)

↳ fibers are spread out

H. Visual cortex

cuneus → sup fiber → inf deffect VF

lingual → infirior fibers → superiors deffect VF



Exactly the same lesion

1. Visual field characteristics

contralat. homo.

hemianopia

congruent

Ask if 1 or 2 eyes if 1 than before chiazma

② respects horizontal / vertical line

③ Vertical : in. or "congruent"